PYTHON DATA ANALYSIS PROJECT REPORT

PROJECT TITLE

Supply Chain Data Analysis Using Python

PROJECT OVERVIEW

This project presents a comprehensive Exploratory Data Analysis (EDA) of a real-world Supply Chain dataset, covering product performance, customer demographics, supplier reliability, inventory, and transportation costs. The analysis begins with rigorous data cleaning to validate completeness and consistency, ensuring the dataset is reliable for downstream analysis and decision-making.

Using Python-based data visualization techniques, the report explores Revenue Analysis, Product Analysis, Supplier Analysis, and Transportation & Logistics Analysis to uncover patterns in profitability, pricing, quality, and delivery performance across the business.

Each visualization is paired with clear business insights and practical recommendations, demonstrating how data-driven decision making can guide strategic actions in sourcing, inventory planning, and logistics optimization.

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Report Type	Python Data Analysis Portfolio Project

PYTHON CONCEPTS DEMONSTRATED

Data Cleaning	Exploratory Data Analysis (EDA)
Pandas	NumPy
Matplotlib	Seaborn
Data Visualization	GroupBy & Aggregation
Statistical Analysis	Correlation Analysis
Business Insights	Business Recommendations

PURPOSE

This project demonstrates practical Python data analysis skills by applying exploratory data analysis, data visualization, and business analytics techniques to solve real-world supply chain problems. It reflects the ability to translate raw data into actionable, business-relevant insights that support informed, data-driven decision making.

Executive Summary & Dataset Overview

This report analyzes a supply chain dataset covering product performance, customer demographics, supplier reliability, inventory, and transportation costs. The dataset was checked for data quality before analysis: no missing values and no duplicate records were found across any column, confirming the data is clean and reliable for decision-making.

100
RECORDS (ROWS)

24
DATA FIELDS (COLUMNS)

0
MISSING VALUES

29
CHARTS & INSIGHTS

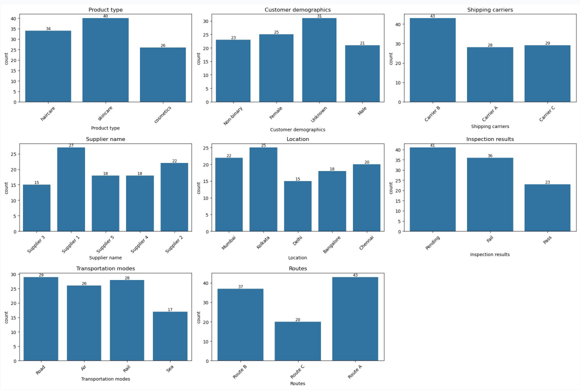
The following pages present each visualization alongside its explanation, the business insight it reveals, and a corresponding recommendation — organized by theme: dataset structure, revenue & profitability, product analysis, supplier analysis, and transportation & shipping cost.

CONTENTS

Countplot Grid — Categorical Columns

SOURCE CODE

```
# List of categorical columns (excluding SKU)
categorical_cols = [
    'Product type',
    'Customer demographics',
    'Shipping carriers',
    'Supplier name',
    'Location',
    'Inspection results',
    'Transportation modes',
    'Routes'
]

# Set up subplots grid
fig, axes = plt.subplots(nrows=3, ncols=3, figsize=(18, 12))
axes = axes.flatten()

# Loop through categorical columns
for i, col in enumerate(categorical_cols):
    ax = axes[i]
    sns.countplot(data=df, x=col, ax=ax)
    ax.set_title(col)
    ax.tick_params(axis='x', rotation=45)

    # Add bar labels
    for container in ax.containers:
        ax.bar_label(container, fontsize=9)

# Hide any unused subplot spaces
for j in range(len(categorical_cols), len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```



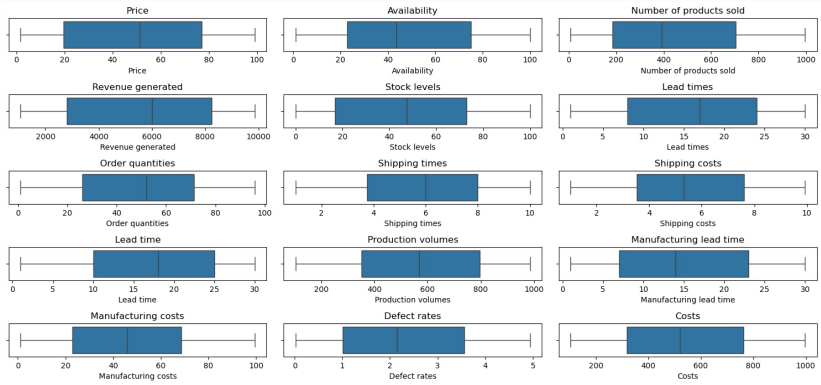
EXPLANATION

- This chart shows the distribution of records across different categories like product type, customers, suppliers, shipping carriers, locations, and routes.
- It helps us understand the overall structure of the dataset before further analysis
- Skincare has the highest number of products (40) followed by Haircare (34) and Cosmetics (26)
- The Unknown customer group has the largest share (31 records)
- Supplier 1 contributes the most records (27)
- Route A is the most frequently used route (43 records) showing higher dependency on one shipping path.

This chart displays boxplots for all numerical columns such as price, revenue, costs, stock levels, and lead times.

SOURCE CODE

```
# Select only numerical columns
num_cols = df.select_dtypes(include='number').columns

# Plot each numerical column in a grid
plt.figure(figsize=(15, 8))
for i, col in enumerate(num_cols, 1):
    plt.subplot(len(num_cols)//3 + 1, 3, i)  # adjust grid size
    sns.boxplot(x=df[col])
    plt.title(col)

plt.tight_layout()
plt.show()
```



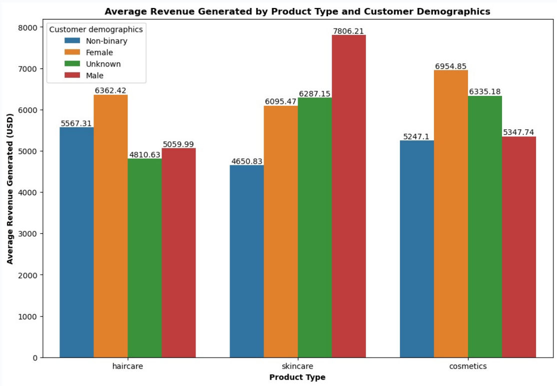
EXPLANATION

- It is used to identify unusual values or outliers before performing further analysis
- The data looks clean, with no invalid values like negative prices or stock levels
- Some variables like revenue, costs, and production volume have a wider range, which is expected because different products have different prices and manufacturing requirements.

This chart shows the average revenue generated by each product type based on different customer groups.

SOURCE CODE

```
#Revenue Generated by each gender with product type
plt.figure(figsize = (12,8))
count = sns.barplot(data = df , x = 'Product type' , y = 'Revenue generated' , hue = 'Customer demographics' , errorbar=None)
count.bar_label(count.containers[0])
count.bar_label(count.containers[1])
count.bar_label(count.containers[2])
count.bar_label(count.containers[3])
plt.title("Average Revenue Generated by Product Type and Customer Demographics" , fontweight = 'bold')
plt.xlabel("Product Type", fontweight = 'bold')
plt.ylabel("Average Revenue Generated (USD)", fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify which customer segment contributes more revenue for specific products
- Male customers generate the highest average revenue from skincare ($7,806) while female customers show the highest average revenue in cosmetics ($6,955)
- Non-binary customers have lower average revenue across product categories compared to other groups

BUSINESS INSIGHT

Customer preferences vary by product type, so targeted marketing strategies can improve sales.

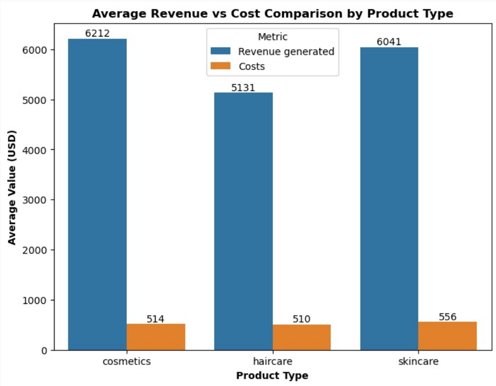
RECOMMENDATION

For example, skincare promotions can focus more on male customers, while cosmetics campaigns can target female customers.

This chart compares the average revenue generated and average cost for each product category.

SOURCE CODE

```
#Which product type (Skincare / Haircare / Cosmetics) generates the most revenue — and is it
proportional to its cost?
rev_cost = df.groupby('Product type')[['Revenue generated','Costs']].mean().reset_index()

rev_cost_melted = rev_cost.melt(id_vars='Product type',
                                value_vars=['Revenue generated','Costs'],
                                var_name='Metric',
                                value_name='Value')

plt.figure(figsize=(8,6))
ax = sns.barplot(data=rev_cost_melted, x='Product type', y='Value', hue='Metric',
errorbar=None)

for container in ax.containers:
    ax.bar_label(container, fmt='%.0f')

plt.title("Average Revenue vs Cost Comparison by Product Type" , fontweight = 'bold')
plt.xlabel("Product Type", fontweight = 'bold')
plt.ylabel("Average Value (USD)", fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify which product type provides better profitability by comparing income with expenses
- Cosmetics performs the best, generating the highest average revenue ($6,212) with the lowest average cost ($514)
- Haircare has the lowest revenue ($5,131) With similar costs, making it less profitable
- Skincare generates good revenue ($6,041) but has slightly higher costs ($556)

BUSINESS INSIGHT

Cosmetics appears to be the most profitable product category due to its strong revenue and lower costs.

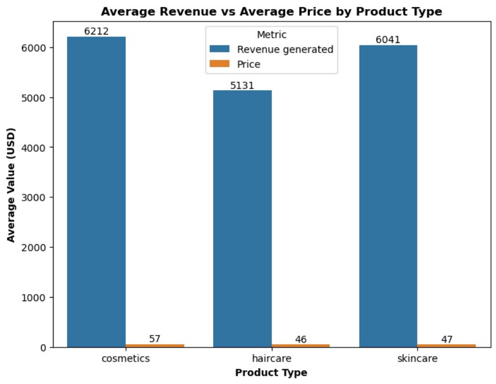
RECOMMENDATION

The company can focus more on cosmetics to improve overall profitability.

This chart compares the average product price with the average revenue generated by each category

SOURCE CODE

```
#Is there a relationship between price and revenue, or do cheaper products sell more and
compensate?
rev_cost = df.groupby('Product type')[['Revenue generated','Price']].mean().reset_index()

rev_cost_melted = rev_cost.melt(id_vars='Product type',
                                value_vars=['Revenue generated','Price'],
                                var_name='Metric',
                                value_name='Value')

plt.figure(figsize=(8,6))
ax = sns.barplot(data=rev_cost_melted, x='Product type', y='Value', hue='Metric',
errorbar=None)

for container in ax.containers:
    ax.bar_label(container, fmt='%.0f')

plt.title("Average Revenue vs Average Price by Product Type",fontweight = 'bold')
plt.xlabel("Product Type", fontweight = 'bold')
plt.ylabel("Average Value (USD)", fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps understand whether higher prices contribute to better revenue performance
- Cosmetics has the highest average price ($57.36) and also generates the highest average revenue ($6,212), showing that higher pricing does not negatively impact sales
- Haircare has the lowest price ($46.01) but also the lowest revenue ($5,131), meaning lower prices did not result in higher earnings.

BUSINESS INSIGHT

The analysis suggests that higher pricing can still generate strong revenue.

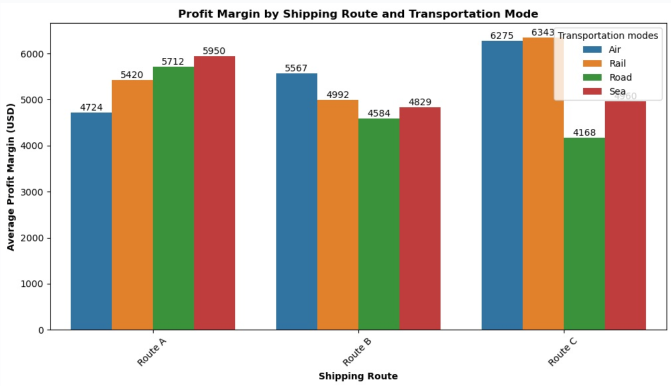
RECOMMENDATION

Businesses should focus on product value and customer demand rather than relying only on lower prices to increase sales.

This chart compares profit margins across different combinations of routes and transportation methods.

SOURCE CODE

```
#Which routes and transportation modes are eating into margins the most?'''
# Calculate margin
df['Margin'] = df['Revenue generated'] - df['Costs']

# Group by Routes and Transportation modes
margin_summary = df.groupby(['Routes','Transportation modes'])['Margin'].mean().reset_index()

# Plot grouped barplot
plt.figure(figsize=(12,6))
ax = sns.barplot(data=margin_summary, x='Routes', y='Margin', hue='Transportation modes',
errorbar=None)

# Add bar labels
for container in ax.containers:
    ax.bar_label(container, fmt='%.0f')

plt.title("Profit Margin by Shipping Route and Transportation Mode" , fontweight = 'bold')
plt.xlabel("Shipping Route" , fontweight = 'bold')
plt.ylabel("Average Profit Margin (USD)" , fontweight = 'bold')
plt.xticks(rotation=45)
plt.show()
```



EXPLANATION

- It helps identify which logistics combinations generate higher profits and which reduce margins.
- Route C with Rail ($6,343) and Air ($6,275) shows the highest average margins while Route C with Road has the lowest margin ($4,168).
- This shows that the same route can perform differently depending on the transportation method used.

BUSINESS INSIGHT

Profitability depends on the combination of route and transport mode not just one factor alone.

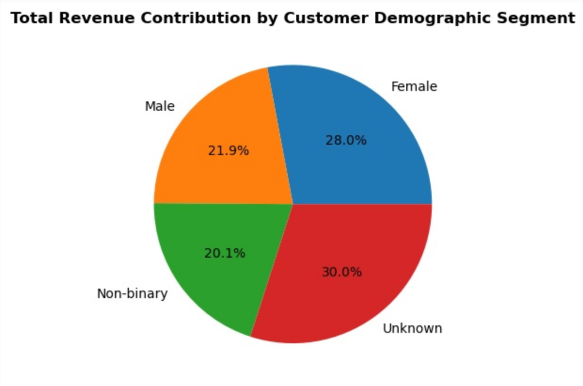
RECOMMENDATION

The company should optimize logistics pairings, such as avoiding less profitable combinations like Route C with Road.

This pie chart shows how much each customer demographic contributes to the company's total revenue

SOURCE CODE

```
revenue_by_demo = df.groupby('Customer demographics')['Revenue generated'].sum()

plt.pie(
    revenue_by_demo.values,
    labels=revenue_by_demo.index,
    autopct='%1.1f%%',
)
plt.title("Total Revenue Contribution by Customer Demographic Segment",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps understand which customer groups are driving overall sales.
- The Unknown customer group contributes the highest revenue share (30%), followed by Female customers (28%).
- Male customers contribute 21.9% and Non-binary customers contribute 20.1% of total revenue.
- The large contribution from unknown customers indicates missing demographic information.

BUSINESS INSIGHT

The large contribution from unknown customers indicates missing demographic information.

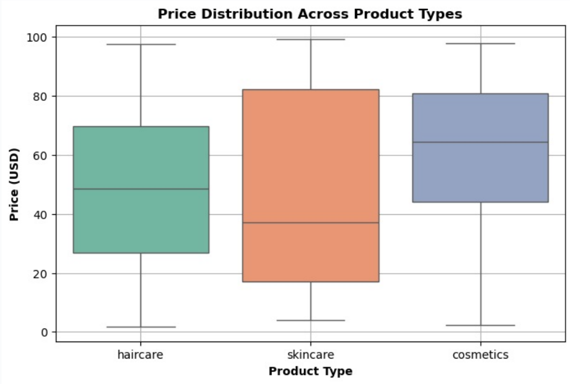
RECOMMENDATION

Improving customer data collection should be a priority because unclear customer profiles make targeted marketing difficult Better demographic tracking can help create more effective marketing strategies and improve customer understanding.

This boxplot shows how product prices vary within each category.

SOURCE CODE

```
#Price Distribution Across Product Types
plt.figure(figsize=(8,5))
sns.boxplot(data=df , x='Product type' , y='Price' , palette = "Set2")
plt.title("Price Distribution Across Product Types" , fontweight = 'bold')
plt.xlabel("Product Type" , fontweight = 'bold')
plt.ylabel("Price (USD)", fontweight = 'bold')
plt.grid()
plt.show()
```



EXPLANATION

- It helps understand the price range,consistency and differences between product types
- Skincare has the widest price range (around $4 to $99), showing that it includes both low-cost and premium products.
- Cosmetics has a more consistent and higher price range, while Haircare falls between the two with moderate price variation.

BUSINESS INSIGHT

Cosmetics has a more consistent and higher price range, while Haircare falls between the two with moderate price variation.

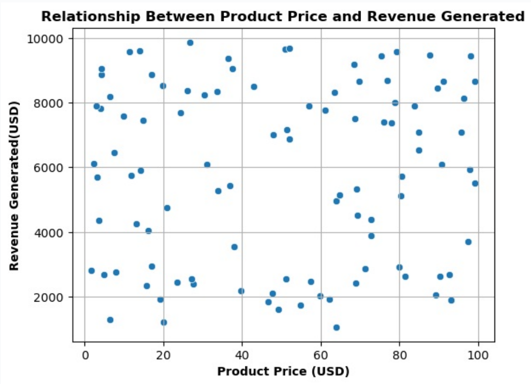
RECOMMENDATION

Skincare has a highly diverse pricing structure, so separating it into budget and premium segments could help improve pricing strategies,inventory planning, and marketing decisions.

This scatter plot shows the relationship between product price and revenue generated

SOURCE CODE

```
#Price vs. Revenue Relationship
sns.scatterplot( data = df , x = 'Price' , y = 'Revenue generated')
plt.grid()
plt.title("Relationship Between Product Price and Revenue Generated",fontweight = 'bold')
plt.xlabel("Product Price (USD)",fontweight = 'bold')
plt.ylabel("Revenue Generated(USD)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps determine whether higher-priced products consistently generate more revenue.
- The points are widely scattered with no clear pattern, and the correlation value (0.038) indicates almost no relationship between price and revenue
- This means that increasing the price of a product does not necessarily lead to higher revenue in this dataset.

BUSINESS INSIGHT

Revenue is influenced by factors beyond price, such as customer demand, marketing, and product popularity.

RECOMMENDATION

The business should focus on improving these areas rather than relying only on price changes to increase revenue.

This hexbin chart shows the relationship between product price and the number of units sold

SOURCE CODE

```
#Is revenue proportional to product price, or do cheaper products compensate through higher
sales volume?
plt.figure(figsize=(8,6))
plt.hexbin(df['Price'], df['Number of products sold'], gridsize=20, cmap='Blues')
plt.colorbar(label='Count of points')
plt.xlabel("Product Price (USD)",fontweight = 'bold')
plt.ylabel("Number of products sold",fontweight = 'bold')
plt.title("Price vs Sales Volume Density (Hexbin)",fontweight = 'bold')
plt.show()
```

EXPLANATION

- Darker areas represent price ranges where more products are concentrated
- The chart shows no clear pattern between lower prices and higher sales volume
- The correlation value (0.006) indicates almost no relationship between price and units sold, meaning cheaper products do not necessarily sell more.

BUSINESS INSIGHT

The correlation value (0.006) indicates almost no relationship between price and units sold, meaning cheaper products do not necessarily sell more.

RECOMMENDATION

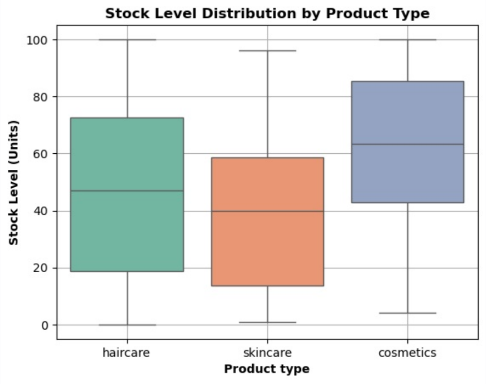
Reducing prices may not significantly increase sales volume in this business Instead, the company should focus on other factors like product quality, customer demand, and marketing strategies to improve sales

PRODUCT ANALYSIS · CHART 1 OF 9

This boxplot shows how inventory levels vary across different product categories

SOURCE CODE

```
#Which product type has the highest stock levels?
#Are we overstocking certain products?
sns.boxplot(data = df , x = 'Product type' , y = 'Stock levels' , palette = 'Set2')
plt.grid()
plt.title("Stock Level Distribution by Product Type",fontweight = 'bold')
plt.xlabel("Product type",fontweight = 'bold')
plt.ylabel("Stock Level (Units)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps compare stock availability and identify possible differences in inventory management
- Cosmetics has the highest average stock level (58.7 units), while Skincare has the lowest (40.2 units)
- Despite having the highest number of products and production volume
- This indicates that inventory levels are not evenly aligned with product demand or production

BUSINESS INSIGHT

The company should review inventory allocation.

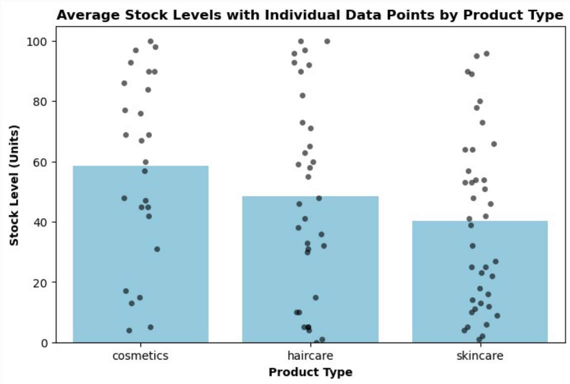
RECOMMENDATION

Cosmetics may have excess stock while Skincare inventory may need better planning to maintain availability and avoid shortages.

This chart shows the average stock level for each product category along with individual product stock values

SOURCE CODE

```
plt.figure(figsize=(8,5))
sns.barplot(data=avg_stock,x="Product type",y="Stock levels",color="skyblue")
sns.stripplot(data=df,x="Product type",y="Stock levels",color="black",alpha=0.6)
plt.title("Average Stock Levels with Individual Data Points by Product Type",fontweight =
'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Stock Level (Units)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps understand both the overall inventory level and the variation between products.
- Cosmetics has a high average stock level, but individual products show large differences — some have very low stock
- While others have excess inventory. Similar variations are also seen in Skincare and Haircare, indicating inconsistent inventory distribution.

BUSINESS INSIGHT

While others have excess inventory. Similar variations are also seen in Skincare and Haircare, indicating inconsistent inventory distribution.

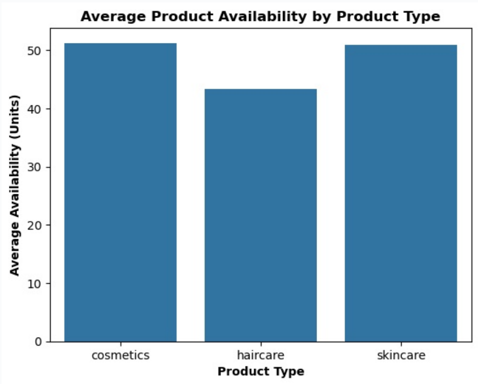
RECOMMENDATION

The company should implement better inventory planning, such as product-wise reorder levels to reduce situations of both stock shortages and excess inventory.

This chart compares the average availability of products across different categories

SOURCE CODE

```
#Which product type has the lowest availability?
low_availability = df.groupby('Product type')['Availability'].mean().reset_index()
sns.barplot(data=low_availability, x='Product type', y='Availability')
plt.title("Average Product Availability by Product Type",fontweight = 'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Average Availability (Units)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps understand how easily customers can access each product type.
- Cosmetics has the highest availability score (51.2), followed closely by Skincare (50.9)
- Haircare has the lowest availability (43.3), even though its stock levels are not the lowest, suggesting other factors may affect product availability.

BUSINESS INSIGHT

Haircare availability issues may not be caused by insufficient stock.

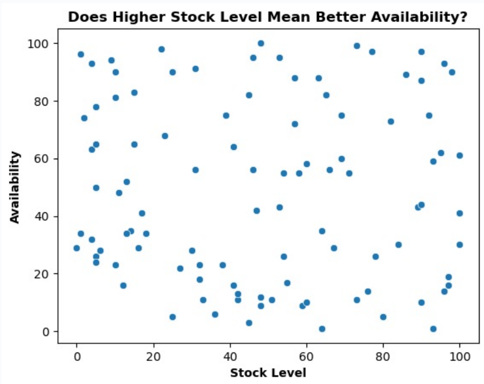
RECOMMENDATION

The company should review processes like distribution, order fulfillment, and inventory handling to improve product accessibility.

This scatter plot shows the relationship between product stock levels and availability scores.

SOURCE CODE

```
sns.scatterplot(data=df,x="Stock levels",y="Availability")
plt.title("Does Higher Stock Level Mean Better Availability?",fontweight = 'bold')
plt.xlabel("Stock Level",fontweight = 'bold')
plt.ylabel("Availability",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps check whether keeping more inventory directly improves product availability
- The points are widely scattered with no clear pattern, and the correlation value (-0.026)
- Shows almost no relationship between stock levels and availability. This means having more stock does not always guarantee better product availability

BUSINESS INSIGHT

Shows almost no relationship between stock levels and availability. This means having more stock does not always guarantee better product availability

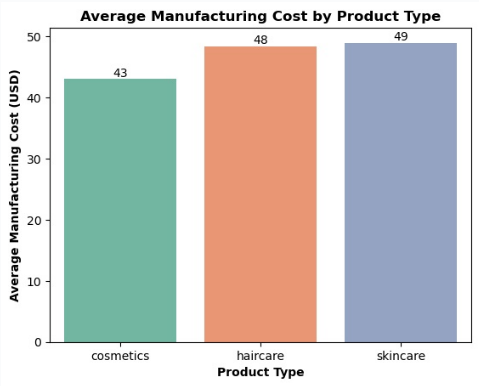
RECOMMENDATION

Improving availability requires focusing on supply chain processes like restocking, inventory management, and demand forecasting rather than simply increasing stock levels.

This chart compares the average cost of producing each product type

SOURCE CODE

```
#Which product type has the highest manufacturing cost?
df_grouped = df.groupby('Product type', as_index=False)['Manufacturing costs'].mean()
ax = sns.barplot(data=df_grouped, x='Product type', y='Manufacturing costs', errorbar=None ,
palette = 'Set2')
for container in ax.containers:
    ax.bar_label(container, fmt='%.0f')
plt.title("Average Manufacturing Cost by Product Type",fontweight = 'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Average Manufacturing Cost (USD)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify which categories are more cost-efficient to manufacture.
- Skincare ($48.99) And Haircare ($48.46) have higher manufacturing costs compared to Cosmetics ($43.05)
- Since Skincare is produced in large quantities, even a small cost difference can significantly impact overall expenses.

BUSINESS INSIGHT

Since Skincare is produced in large quantities, even a small cost difference can significantly impact overall expenses.

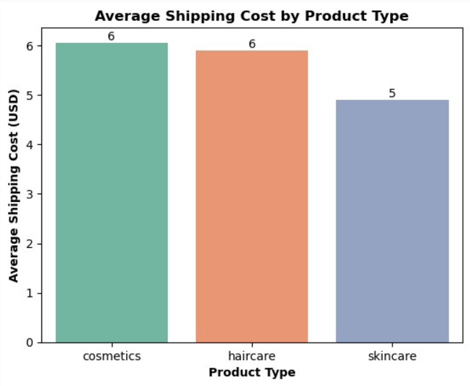
RECOMMENDATION

Cosmetics is the most cost-efficient category because it has lower manufacturing costs and higher revenue performance The company can consider giving more focus to cosmetics production for better profitability.

This chart compares the average shipping cost for each product category

SOURCE CODE

```
#Which product type has the highest shipping cost
high_shippingcost = df.groupby('Product type')['Shipping costs'].mean().reset_index()
ax = sns.barplot(data = high_shippingcost , x = 'Product type' , y = 'Shipping costs' , palette
= 'Set2' , errorbar = None)
for container in ax.containers:
    ax.bar_label(container, fmt='%.0f')
plt.title("Average Shipping Cost by Product Type",fontweight = 'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Average Shipping Cost (USD)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps understand how transportation expenses affect the overall cost of products.
- Cosmetics has the highest shipping cost ($6.06 per unit) while Skincare has the lowest ($4.91)
- However, despite higher shipping expenses, Cosmetics remains the most profitable category due to its lower manufacturing cost and stronger revenue performance.

BUSINESS INSIGHT

However, despite higher shipping expenses, Cosmetics remains the most profitable category due to its lower manufacturing cost and stronger revenue performance.

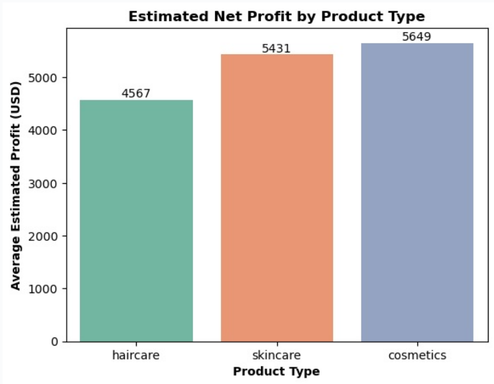
RECOMMENDATION

The company should monitor cosmetics shipping costs and explore ways to reduce transportation expenses but these costs currently do not significantly impact its profitability advantage.

This chart shows the estimated profit for each product category after considering all major costs

SOURCE CODE

```
#Which product type appears most profitable based on estimated profit?
df["Estimated Profit"] = (df["Revenue generated"]- df["Manufacturing costs"]- df["Shipping
costs"]- df["Costs"])
product_profit = df.groupby('Product type')['Estimated Profit'].mean()
ax = sns.barplot(data = df , x = 'Product type' , y = 'Estimated Profit' , palette = 'Set2' ,
errorbar = None)
for container in ax.containers:
    ax.bar_label(container, fmt='%.0f')
plt.title("Estimated Net Profit by Product Type",fontweight = 'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Average Estimated Profit (USD)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- Including manufacturing, shipping, and other expenses. It helps identify the most profitable product line
- Cosmetics generates the highest estimated profit ($5,649 per product) followed By Skincare ($5,431) and Haircare ($4,567)
- This confirms that Cosmetics performs best not only in revenue generation but also in overall profitability.

BUSINESS INSIGHT

Cosmetics is the strongest-performing category, making it a suitable area for increased investment and production focus.

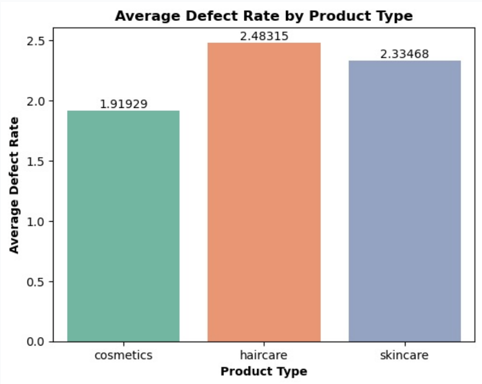
RECOMMENDATION

Haircare may require further analysis to identify ways to improve profitability.

This chart compares the average defect rate across different product categories

SOURCE CODE

```
#Which product type has the highest defect rate
avg_defect = df.groupby('Product type')['Defect rates'].mean().reset_index()
ax = sns.barplot(data = avg_defect , x = 'Product type' , y = 'Defect rates' , palette = 'Set2'
, errorbar = None)
for container in ax.containers:
    ax.bar_label(container)
plt.title("Average Defect Rate by Product Type",fontweight = 'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Average Defect Rate",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify which product line has better quality control and fewer product issues.
- Cosmetics has the lowest defect rate (1.92%), indicating better quality performance.
- Haircare has the highest defect rate (2.48%), suggesting a higher chance of quality-related issues
- This matches earlier findings where Haircare also showed lower revenue and profitability.

BUSINESS INSIGHT

Cosmetics performs strongly in both profitability and quality.

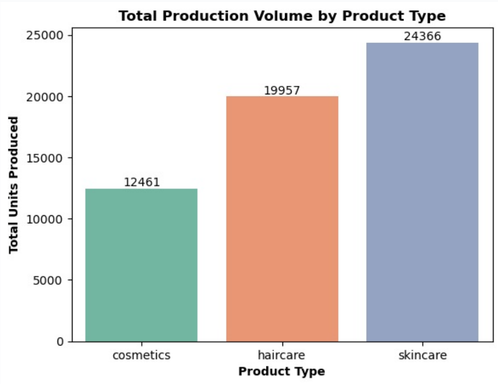
RECOMMENDATION

Haircare requires further investigation to identify whether production processes, materials, or quality control issues are affecting its overall performance.

This chart shows the total number of units produced for each product category.

SOURCE CODE

```
#Which product contributes the most to total production volume
total_production = df.groupby('Product type')['Production volumes'].sum().reset_index()
ax = sns.barplot(data = total_production , x = 'Product type' , y = 'Production volumes' ,
palette = 'Set2' , errorbar = None)
for container in ax.containers:
    ax.bar_label(container)
plt.title("Total Production Volume by Product Type",fontweight = 'bold')
plt.xlabel("Product Type",fontweight = 'bold')
plt.ylabel("Total Units Produced",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps understand where the company's manufacturing capacity is being utilized.
- Skincare has the highest production volume (24,366 units), followed by Haircare (19,957) and Cosmetics (12,461)
- However, earlier analysis showed that Cosmetics generates higher profit despite having the lowest production volume.

BUSINESS INSIGHT

However, earlier analysis showed that Cosmetics generates higher profit despite having the lowest production volume.

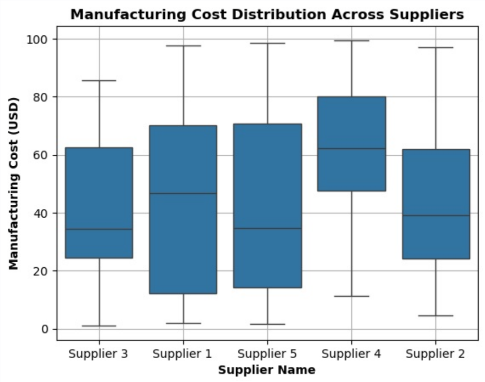
RECOMMENDATION

The company may need to reconsider its production allocation Increasing focus on Cosmetics production while optimizing Skincare output could help improve overall profitability.

This boxplot compares manufacturing costs across different suppliers

SOURCE CODE

```
#Which supplier has the highest manufacturing cost?
#Which supplier is the most expensive for manufacturing?
sns.boxplot(data=df, x="Supplier name", y="Manufacturing costs")
plt.title("Manufacturing Cost Distribution Across Suppliers",fontweight = 'bold')
plt.xlabel("Supplier Name",fontweight = 'bold')
plt.ylabel("Manufacturing Cost (USD)",fontweight = 'bold')
plt.grid()
plt.show()
```



EXPLANATION

- It helps identify suppliers with unusually high or low production costs.
- Supplier 4 has the highest average manufacturing cost ($62.71 per unit) while the other suppliers remain around $42–45 per unit
- This significant cost difference makes Supplier 4 a clear outlier compared to the others.

BUSINESS INSIGHT

This significant cost difference makes Supplier 4 a clear outlier compared to the others.

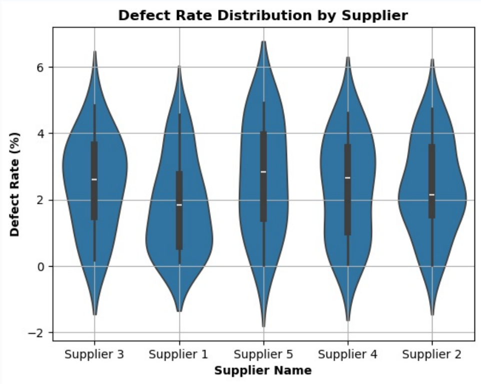
RECOMMENDATION

The company should review Supplier 4’s pricing and negotiate better rates, especially if the higher cost does not provide additional benefits like improved quality or performance.

This violin plot compares defect rates across different suppliers

SOURCE CODE

```
#Which supplier delivers products with the poorest quality?
sns.violinplot(data=df,x="Supplier name",y="Defect rates")
plt.title("Defect Rate Distribution by Supplier",fontweight = 'bold')
plt.xlabel("Supplier Name",fontweight = 'bold')
plt.ylabel("Defect Rate (%)",fontweight = 'bold')
plt.grid()
plt.show()
```



EXPLANATION

- It helps identify which suppliers maintain better product quality and which have more quality issues.
- Supplier 1 has the lowest average defect rate (1.80%), showing the best quality performance
- Supplier 5 has the highest defect rate (2.67%), indicating more frequent quality issues compared to other suppliers.

BUSINESS INSIGHT

Supplier 5 has the highest defect rate (2.67%), indicating more frequent quality issues compared to other suppliers.

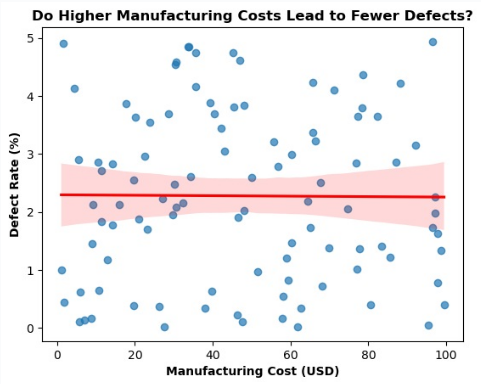
RECOMMENDATION

Supplier 1 can be used as a quality benchmark, while suppliers with higher defect rates should improve their quality control processes to reduce defects and maintain consistent standards.

This regression plot shows the relationship between manufacturing cost and defect rate

SOURCE CODE

```
#Are suppliers charging more because they produce higher-quality products?
sns.regplot(data=df,x="Manufacturing costs",y="Defect rates",scatter_kws=
{"alpha":0.7},line_kws={"color":"red"})
plt.title("Do Higher Manufacturing Costs Lead to Fewer Defects?",fontweight = 'bold')
plt.xlabel("Manufacturing Cost (USD)",fontweight = 'bold')
plt.ylabel("Defect Rate (%)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps check whether paying higher manufacturing costs results in better product quality.
- The trend line is almost flat, and the correlation value (-0.008) shows no meaningful relationship between cost and defect rate
- This means higher manufacturing costs do not guarantee fewer defects.

BUSINESS INSIGHT

This means higher manufacturing costs do not guarantee fewer defects.

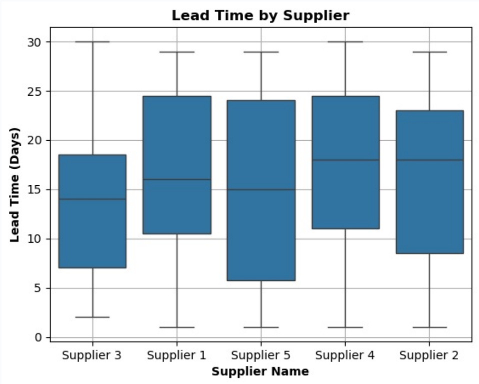
RECOMMENDATION

The company should evaluate suppliers based on both cost and quality performance Higher-cost suppliers, like Supplier 4, should be reviewed to ensure the extra cost provides real value.

This boxplot compares the delivery time of different suppliers

SOURCE CODE

```
#Which supplier takes the longest to deliver manufactured products?
sns.boxplot(data=df,x="Supplier name",y="Lead times")
plt.title("Lead Time by Supplier",fontweight = 'bold')
plt.xlabel("Supplier Name",fontweight = 'bold')
plt.ylabel("Lead Time (Days)",fontweight = 'bold')
plt.grid()
plt.show()
```



EXPLANATION

- It helps understand which suppliers provide faster and more consistent deliveries.
- Supplier 4 has the highest average lead time (17.0 days), making it the slowest supplier, while Supplier 3 is the fastest (14.3 days)
- Along with higher manufacturing costs and no clear quality advantage, Supplier 4 shows weaker overall performance.

BUSINESS INSIGHT

Along with higher manufacturing costs and no clear quality advantage, Supplier 4 shows weaker overall performance.

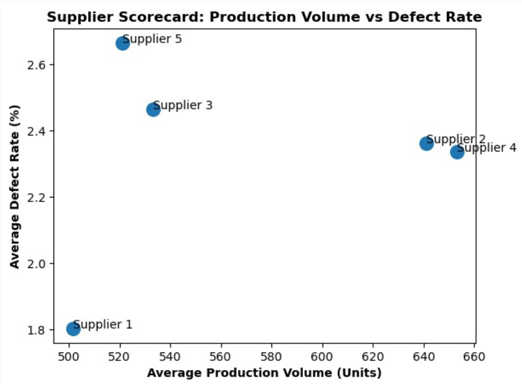
RECOMMENDATION

The company should review its relationship with Supplier 4, as it performs poorly in cost, delivery speed, and quality Reducing dependency on underperforming suppliers can improve supply chain efficiency.

This scatter plot compares suppliers based on their production volume and defect rate

SOURCE CODE

```
#Which supplier combines high production with low defect rates?
supplier_summary = df.groupby("Supplier name").agg({"Production volumes":"mean","Defect
rates":"mean"}).reset_index()
sns.scatterplot(data=supplier_summary,x="Production volumes",y="Defect rates",s=180)
for i in range(len(supplier_summary)):
    plt.text(
        supplier_summary["Production volumes"][i],
        supplier_summary["Defect rates"][i],
        supplier_summary["Supplier name"][i]
    )
plt.title("Supplier Scorecard: Production Volume vs Defect Rate",fontweight = 'bold')
plt.xlabel("Average Production Volume (Units)",fontweight = 'bold')
plt.ylabel("Average Defect Rate (%)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify suppliers that provide high output while maintaining good quality.
- Supplier 1 shows the best overall performance with strong production volume and the lowest defect rate
- Supplier 5 performs the weakest, with lower production volume and the highest defect rate among all suppliers.

BUSINESS INSIGHT

Supplier 5 performs the weakest, with lower production volume and the highest defect rate among all suppliers.

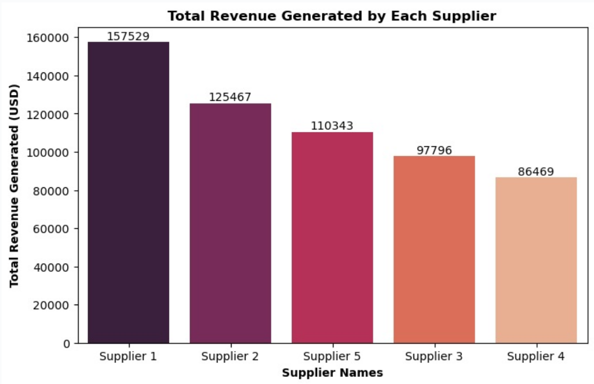
RECOMMENDATION

Supplier 1 can be considered for increased business allocation, while Supplier 5 should be reviewed for improvement or possible replacement to improve overall supply chain performance.

This chart shows the total revenue generated from products supplied by each supplier

SOURCE CODE

```
#Which supplier generates the highest revenue?
supplier_revenue = (df.groupby("Supplier name")["Revenue
generated"].sum().sort_values(ascending=False).reset_index())
plt.figure(figsize=(8,5))
ax = sns.barplot(data=supplier_revenue,x="Supplier name",y="Revenue
generated",palette="rocket")
for container in ax.containers:
    ax.bar_label(container)
plt.title("Total Revenue Generated by Each Supplier",fontweight = 'bold')
plt.xlabel("Supplier Names",fontweight = 'bold')
plt.ylabel("Total Revenue Generated (USD)",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify which supplier relationships contribute the most to overall business revenue.
- Supplier 1 generates the highest total revenue ($157,529), while Supplier 4 contributes the lowest ($86,469).
- This matches previous findings where Supplier 1 performed better
- In quality, cost efficiency, and delivery speed, while Supplier 4 showed weaker performance.

BUSINESS INSIGHT

In quality, cost efficiency, and delivery speed, while Supplier 4 showed weaker performance.

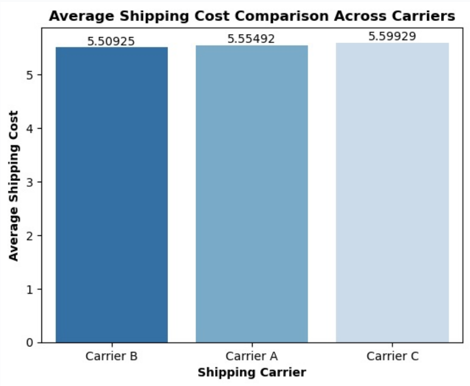
RECOMMENDATION

Supplier 1 appears to be the strongest overall partner and can be considered for increased business allocation Supplier 4 should be reviewed and renegotiated due to its higher costs, slower delivery, and lower business contribution.

This chart compares the average shipping costs of different carriers

SOURCE CODE

```
#Which shipping carrier has the lowest shipping cost?
carrier_cost = (df.groupby("Shipping carriers")["Shipping
costs"].mean().sort_values().reset_index())
ax = sns.barplot(data=carrier_cost,x="Shipping carriers",y="Shipping costs",palette="Blues_r")
for container in ax.containers:
    ax.bar_label(container)
plt.title("Average Shipping Cost Comparison Across Carriers",fontweight = 'bold')
plt.xlabel("Shipping Carrier",fontweight = 'bold')
plt.ylabel("Average Shipping Cost",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps determine whether one carrier provides a significant cost advantage over others.
- All three carriers have very similar shipping costs, with Carrier B being slightly cheaper ($5.51),
- Followed by Carrier A ($5.55) and Carrier C ($5.60). The difference is very small, indicating that cost is not a major factor when choosing a carrier.

BUSINESS INSIGHT

Followed by Carrier A ($5.55) and Carrier C ($5.60). The difference is very small, indicating that cost is not a major factor when choosing a carrier.

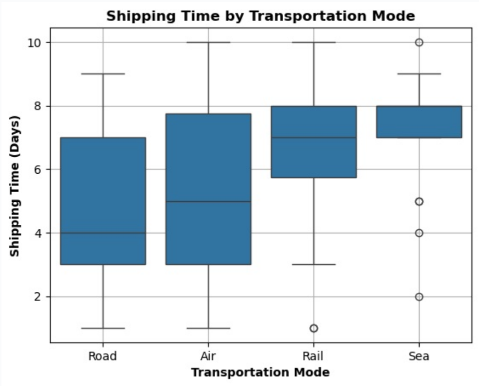
RECOMMENDATION

Since shipping costs are nearly the same across carriers, the company should focus on factors like delivery reliability, speed, and service quality rather than switching carriers only to reduce costs.

This boxplot compares delivery times across different transportation modes

SOURCE CODE

```
#Which transportation mode is the fastest?
transport_time = (df.groupby("Transportation modes")["Shipping
times"].mean().sort_values().reset_index())
sns.boxplot(data=df,x="Transportation modes",y="Shipping times")
plt.title("Shipping Time by Transportation Mode",fontweight = 'bold')
plt.xlabel("Transportation Mode",fontweight = 'bold')
plt.ylabel("Shipping Time (Days)",fontweight = 'bold')
plt.grid()
plt.show()
print(transport_time)
```



EXPLANATION

- It helps identify which shipping methods are faster and more suitable for different business needs.
- Road is the fastest transportation mode with an average shipping time of 4.7 days, while Sea is the slowest at 7.1 days
- Air (5.1 days) and Rail (6.6 days) fall between these two options.

BUSINESS INSIGHT

Air (5.1 days) and Rail (6.6 days) fall between these two options.

RECOMMENDATION

For urgent deliveries, Road and Air transportation are better choices due to faster delivery times Sea transport can be used for non-urgent shipments where lower cost is more important than speed.

TRANSPORTATION & SHIPPING · CHART 3 OF 4

This chart analyzes the relationship between shipping cost and delivery time

SOURCE CODE

```
##.Does paying higher shipping costs reduce shipping time?
sns.lineplot(data=df, x="Shipping costs", y="Shipping times", estimator="mean",marker="o")
plt.title("Average Shipping Time by Shipping Cost",fontweight = 'bold')
plt.xlabel("Shipping Cost",fontweight = 'bold')
plt.ylabel("Average Shipping Time",fontweight = 'bold')
plt.show()
```

EXPLANATION

- It helps determine whether paying higher shipping costs results in faster deliveries.
- The chart shows no clear pattern between cost and delivery time, and the correlation value (0.045) indicates almost no relationship
- This means higher shipping expenses do not necessarily lead to faster deliveries.

BUSINESS INSIGHT

This means higher shipping expenses do not necessarily lead to faster deliveries.

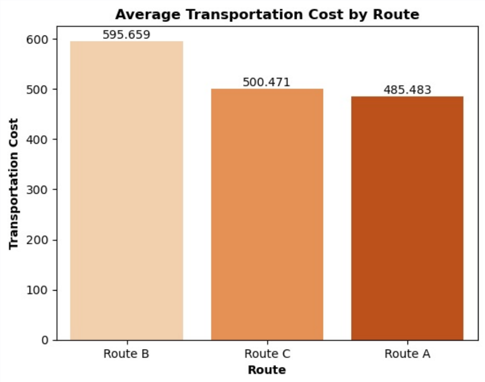
RECOMMENDATION

Faster delivery depends more on the choice of transportation mode rather than spending more on shipping The company should optimize transport selection instead of increasing shipping costs to improve delivery speed.

This chart compares the average transportation cost for each delivery route

SOURCE CODE

```
route_cost = (df.groupby("Routes")["Costs"].mean().sort_values(ascending=False).reset_index())
ax = sns.barplot(data=route_cost,x="Routes",y="Costs",palette="Oranges")
for container in ax.containers:
    ax.bar_label(container)
plt.title("Average Transportation Cost by Route",fontweight = 'bold')
plt.xlabel("Route",fontweight = 'bold')
plt.ylabel("Transportation Cost",fontweight = 'bold')
plt.show()
```



EXPLANATION

- It helps identify which routes are more expensive and where cost optimization opportunities exist.
- Route B has the highest average transportation cost ($595.66), while Route A is the most cost effective route ($485.48)
- The difference of around $110 per shipment can create a significant impact when scaled across many deliveries.

BUSINESS INSIGHT

The difference of around $110 per shipment can create a significant impact when scaled across many deliveries.

RECOMMENDATION

The company should analyze Route B for cost reduction opportunities and consider shifting more shipments toward lower-cost routes like Route A where operationally possible.